

PROFESSIONAL SUMMARY

Pipeline developer and technical director with approximately 10 years in animation and VFX. Develops Python and Maya integrations for asset management, publishing, data extraction, and production automation. Works across departments to translate artist requirements into practical tools and resolve complex pipeline failures across Linux and Windows. Experience includes USD workflows, Flow Production Tracking (ShotGrid), distributed render/compute environments, Deadline management (AWS Deadline Cloud and local repository) and Git CI/CD. Currently building tools that turn Maya scene data into datasets for ML.

TECHNICAL SKILLS

DEVELOPMENT	PYTHON MAYAPY MEL OPENMAYA QT/PYSIDE GIT PYTEST MAXSCRIPT
DCCS & TOOLS	MAYA HOUDINI NUKE 3DS MAX BIFROST XGEN MASH ZIVA GOLAEEM CROWD SOLARIS PDG ARNOLD
DATA & PIPELINES	USD ALEMBIC FLOW PRODUCTION TRACKING (SHOTGRID) PYBLSH ASSET VALIDATION AND PUBLISHING PANDAS PARQUET
INFRASTRUCTURE	DEADLINE (AWS DEADLINE CLOUD) REZ LINUX WINDOWS CI/CD (GITLAB, GITHUB ACTIONS)

PROFESSIONAL EXPERIENCE

PIXOMONDO 2022 - 2026	LEAD SUPPORT DEVELOPER - Led technical investigation and resolution of complex software, pipeline, infrastructure, and production issues. - Developed Python tools and automation to solve production problems and improve user workflows. - Debugged issues spanning application code, APIs, package environments, operating systems, storage and render infrastructure. - Performed code reviews, provided technical guidance, and coordinated technical work across the support team. - Partnered with developers, infrastructure engineers, systems teams, and production users on long-term solutions. SENIOR PIPELINE DEVELOPER - Developed and maintained Python-based production tools used across multiple departments. - Designed and implemented components of the studio's USD production architecture, including department and shot layers. - Developed the initial FX production pipeline, shader workflows, and associated automation. - Developed and maintained integrations with Rez-based package, dependency, and environment management systems. - Used Git/GitLab for source control, collaboration, code review, CI/CD workflows, and software delivery.
SNOWBALL STUDIOS 2019 - 2022	CG SUPERVISOR - Set guidelines for artists to work within a 3D production. - Negotiated and managed clients to set achievable goals. - Worked with department heads to approve deliverables. PIPELINE DEVELOPER - Worked in an Agile/Scrum development team to design, implement, and maintain production software. - Developed Python tools and applications supporting multiple 3D production departments. - Designed the architecture for a task-validation system used to validate assets before downstream processing. - Designed and implemented production pipelines for Crowds, FX, and Edit/Ingest workflows. - Automated asset import, validation, processing, and publishing workflows. - Gathered requirements from users and translated production needs into software solutions.
KEYFRAME DIGITAL 2016 - 2018	PIPELINE DEVELOPER / RIGGING TD - Developed tools and automation for production workflows, primarily using Python and 3ds Max. - Designed and implemented an asset-management pipeline from asset creation through downstream shot departments. - Built internal tools to automate repetitive artist, rigging, and asset-processing workflows. - Supported production users and resolved technical issues involving tools, assets, and workflows.

EDUCATION

2013 - 2016	HUMBER COLLEGE - DIPLOMA - 3D ANIMATION, ART AND DESIGN
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PROJECTS

STUDIO SYSTEMS (case studies on arsamali.com)

- Maya asset representation system:** custom plug-in node and commands that switch each asset between proxy, GPU cache, Alembic, USD and render standins per department, with a Qt asset manager.
- Maya & USD scene I/O:** chunked parallel Alembic, USD export with custom mayaUSD chasers, scene rebuilds from USD, and generated asset and shot stages.
- Playblast, review & QC renders:** Pyblish playblast system with color management and a queue for Nuke, ShotGrid Versions and RV; automated Arnold/V-Ray QC renders on Deadline.

OPEN SOURCE (github.com/arsamasif)

- maya-scene-harvest:** extracts mesh, rig, material and dependency data from folders of Maya and USD scenes into Parquet datasets, with parallel workers, checkpoint/resume, charts and a desktop app.
- maya-rep-switch:** custom Maya transform node (API) for switching asset representations, with undo, a panel and a right-click menu.
- maya-asset-gate:** validate, fix and publish Maya assets to USD with LOD variants and materials; runs inside or outside Maya.
- maya-rig-kit** and **maya-slash-trail:** rigging tools (skin-weight proxies, IK/FK matching, rig checks) and animated slash trails, with Qt UIs. All tested with pytest under mayapy, with CI.